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Iterative Refinement of Point Correspondences for 3D Statistical Shape Models

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Introduction

- Shape atlases : Representation and Modeling
- Bone Atlases for the study of anatomical variation
- Dense Point based representations from registration to a template
- Incorrect point correspondences can influence the statistical modeling



Prior Work

- Active Shape Models

Cootes et al., 1995
Active shape models - their training and application.

- Registration to mean shape

Chui et al., 2004
Unsupervised learning of an atlas from unlabeled point-sets.

- Iterative Bootstrapping

Chintalapani et al., 2007
Statistical Atlases of Bone Anatomy: Construction, Iterative Improvement and Validation

- Optimization of MDL

Davies et al., 2010
Building 3-D Statistical Shape Models by Direct Optimization



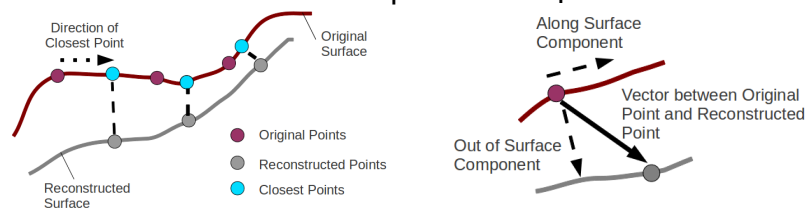
Proposed Method

Iterative method with two steps:

- Use current point correspondences to generate model (PCA)

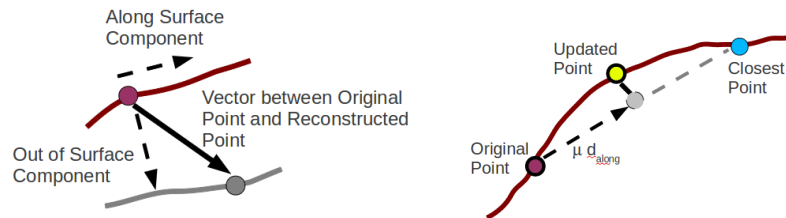
$$V_i^{rec} = \bar{M} + \sum_{i=1}^n \lambda_i Y_i$$

- Use model to re-estimate point correspondences



Fractional Update

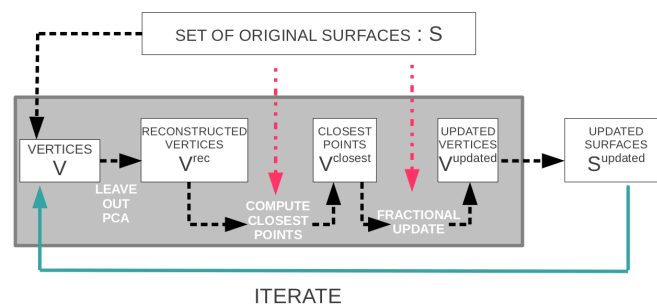
- Closest Points may cause illegal triangles
- Comparing surface normals to check for “flipping” triangles can identify this
- Fractional update: Select largest μ such that no triangles are flipped



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Flowchart



$S = \{V, T\}$: Original Surfaces (Meshes and Triangles)

V : Original Vertices

V^{rec} : Reconstructed Vertices

$V^{closest}$: Closest Points on Original Surface to Reconstructed Vertices

$V^{updated}$: Vertices after fractional update

$S^{updated} = S\{V^{updated}, T\}$: Updated Surfaces

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Algorithm

Algorithm 1 Point Update Algorithm

```
1: Input:  $\mathcal{S} = \{(V_i, T) | i = 1 \dots N\}$ , Stepsize  $s$ 
2: for all  $S_i = (V_i, T) \in \mathcal{S}$  do
3:   Generate  $\mathcal{V}'_i$  and compute PCA model  $\Rightarrow (\bar{M}, Y)$ .
4:   Reconstruct  $V_i$  with  $\bar{M}$  and  $Y \Rightarrow V_i^{rec}$ 
5:   Compute closest points to  $S_i$ .  $\Rightarrow V_i^{closest}$ 
6:   Compute direction vector  $d_{along}$  for every vertex
7:   Set  $\mu = 1$  and compute fractional update  $\Rightarrow V_i^{updated}$ 
8:   while  $V_i^{updated}$  is not consistent do
9:      $\mu = \mu - s$ 
10:    Compute fractional update  $V_i^{updated}$ 
11:  end while
12: end for
13: Output:  $\mathcal{S}^{updated} = \{(V_i^{updated}, T) | i = 1 \dots N\}$ 
```



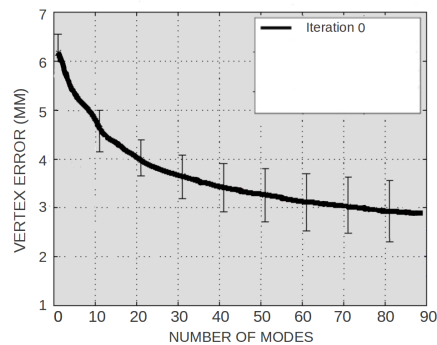
Experimental Setup: Data

- Pelvis Atlas: 110 normal adult male subjects, anonymized
- Each sample is a 512X512X256 CT Volume, manually segmented
- Surface mesh extraction: 11163 points, 23414 triangles
- Each mesh was first registered to a template mesh to establish initial correspondences



Experimental Setup: Validation

- Cross validation: 5 fold
- Training set is used for learning mean shape and modes



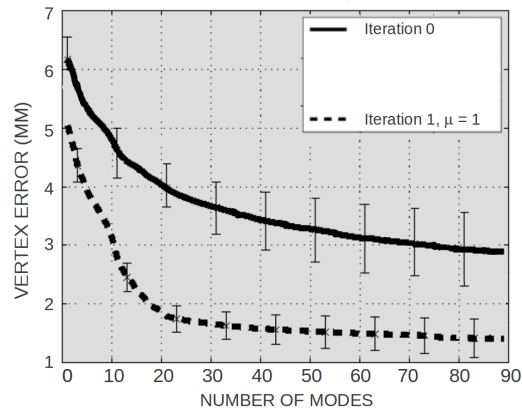
Experimental Setup: Validation

- Cross validation: 5 fold
- Training set is used for learning mean shape and modes
- Reconstruction for all the data is carried out with 30 modes
- Point Update carried out to eliminate out of surface component
- Fractional update carried out to ensure consistency
- Reported Metrics: Vertex Error, Surface Error, Volume Error

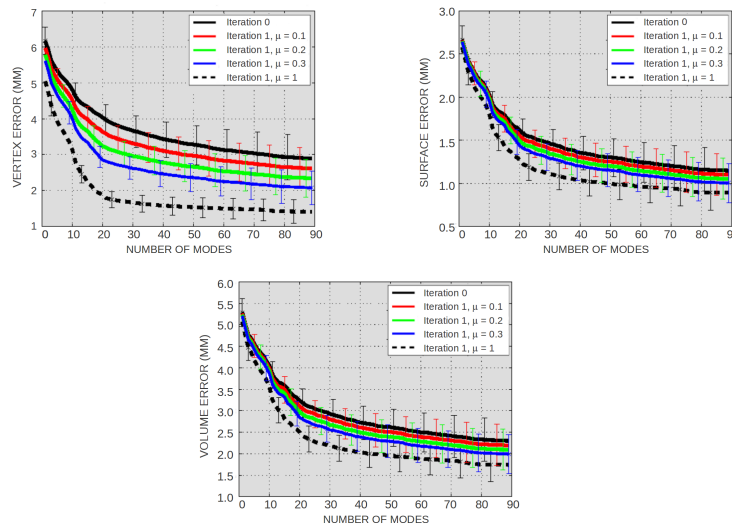


Closest Points

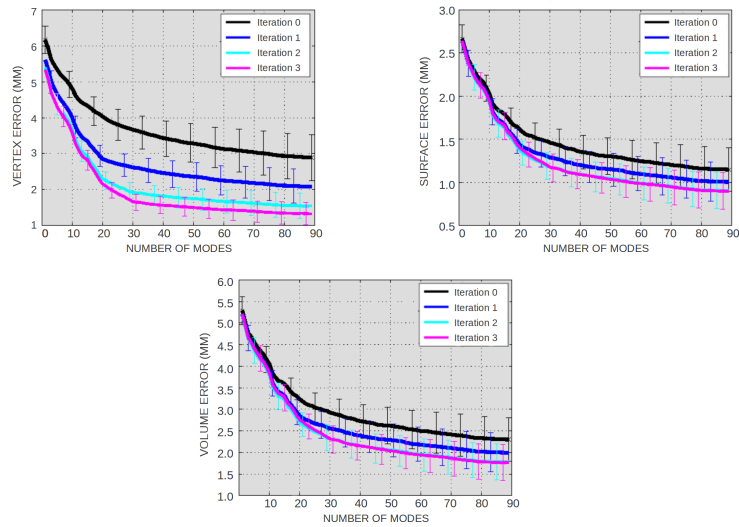
- Leave out Validation with Point Update to the closest points V^{closest}



The effect of Fractional Updates



Multiple Iterations

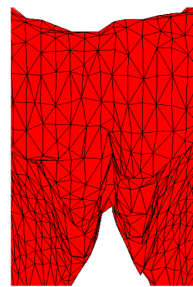


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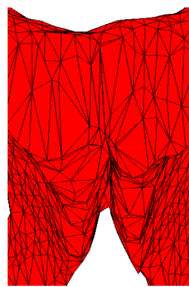
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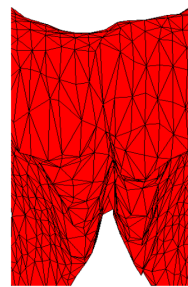
Example of Mesh Update



Target Shape



Iteration 1 –
Closest Points



Iteration 3 – Adaptive
Fractional Update

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Conclusion

- Presented a general iterative method for refinement of 3D point to point correspondences, which does not require any outside information
- Results show reduction in vertex and surface error for leave 20 out validation
- Extensions: Evaluation of consistency using intensity information, application to other types of atlases.



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Questions?

